Victorine Younang, Ph.D.

AI Security & Safety Researcher — Ph.D., Computer Science

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SUMMARY

Machine Learning Researcher (Ph.D., Computer Science) specializing in LLM evaluation, adversarial robustness, and agentic AI security. Expert in automated red-teaming harnesses, execution-verified alignment (DPO), and custom safety metrics for foundation models; identifies AI supply-chain vulnerabilities and neural backdoors via neuro-symbolic static analysis (CipherGuard). Bridges theoretical ML with scalable security infrastructure and GDPR/HIPAA-grade data protection.

TECHNICAL SKILLS

- **AI Safety & Security:** LLM evaluation & safety/robustness metrics, red-teaming, adversarial ML, backdoor/sleeper-agent detection, deception probing, interpretability (SHAP/LIME), formal verification (Isabelle/HOL), threat modeling (CVSS), OWASP Top 10 for LLMs
- **LLMs & Alignment:** fine-tuning (LoRA/QLoRA, MoE), DPO/RLHF, RAG, agentic systems (LangGraph), vLLM; PyTorch, Hugging Face, GNN/GAT
- **Security & Privacy Infra:** HIPAA/PII-safe NLP, GDPR data sanitization, anomaly/fraud/DDoS detection, secure data architecture; Hadoop, Kudu, Elasticsearch, Snowflake
- **Data & Languages:** Python, SQL, R, PySpark; Docker, MLflow

EXPERIENCE

Oakland University | AI Security & Safety Researcher

Michigan, USA · Sep 2020 – Apr 2026

- **Adversarial Robustness & Vulnerability Evaluation:** automated LLM evaluation harness assessing adversarial robustness and semantic consistency of six open frontier models (incl. DeepSeek-V2 MoE, Qwen2.5).
- **Custom Safety Guardrails:** execution-verified safety metrics (syntactic validity, semantic match, value consistency) catching failure modes, prompt-injection, and hallucination triggers missed by BLEU/CodeBLEU.
- **Neural Backdoor & Supply-Chain Detection (CipherGuard):** reference-free static-analysis pipeline using Graph Attention Networks over AST graphs; detected branchless key-leak backdoors missed by constant-time analyzers (NDSS 2027 submission).
- **Model Alignment (RLHF/DPO) & Policy:** curriculum fine-tuning stack with Direct Preference Optimization on constrained hardware; primary researcher on the NSF "Generative AI for Secure Systems" project.

Health Alliance Plan (HAP) | AI Safety & Interpretability Researcher

Detroit, MI · Summer 2025

- **Trustworthy AI & Algorithmic Fairness:** rigorous bias audits and fairness evaluations across demographic subsets (age, zip) so clinical models don't propagate systemic disparities.
- **Interpretability & Transparency:** glass-box frameworks (SHAP, LIME) over deep classifiers → feature-level risk explanations for clinical care managers.
- **Secure & Privacy-Preserving NLP:** HIPAA-compliant pipeline extracting sensitive unstructured text from Epic EHR; secure Snowflake warehouses for anonymized feature generation.
- **Semantic Robustness & NER:** ClinicalBERT NER engineered to handle semantic negations and adversarial shorthand in medical prose.

Rec Room | Applied ML Researcher (Trust & Safety)

Seattle, WA · Summer 2023

- **Structural Anomaly & Abuse Detection:** structural representation model (random walk, Node2Vec) over 1,000+ environments, laying the foundation for detecting coordinated inauthentic behavior and adversarial user clustering.
- **High-Throughput Safety Inference:** refactored the real-time Python pipeline, –10% latency, so anomaly/safety-scoring models run within production latency budgets.
- **Product Guardrails:** deployed the graph engine to production (+33% session time) while maintaining platform integrity and safety guardrails.

EDF (Électricité de France) | R&D ML Researcher (Risk & Probabilistic Modeling)

Paris, France · Summer 2021

- **Tail-Risk & Stress Analysis:** Monte Carlo simulations across extreme spatio-temporal scenarios → quantifiable uncertainty bounds for infrastructure failure.
- **Probabilistic Deep Learning:** quantile-loss sequence models replacing fragile point-estimates with robust 10th–90th percentile intervals for grid stability.
- **Physics-Guided Regularization:** ST-GNNs in PyTorch enforcing operational limits and Kirchhoff's laws via loss penalties.
- **Trustworthy AI:** SHAP feature-importance translating opaque GNN topology into transparent insights for human grid operators.

EDF (Électricité de France) | Data Privacy & AI Security Engineer

Paris, France · 2019–2020

- **Large-Scale Data Sanitization & PII Protection:** secure data infrastructure on distributed Hadoop, executing GDPR data-erasure across millions of sensitive customer profiles to prevent PII leakage into downstream ML.
- **Automated Compliance:** compliance + validation pipelines mitigating privacy risks and reducing downstream processing errors by 30%.

Orange Telecom | Network Anomaly & Data Security Researcher

Paris, France & Brussels, Belgium · 2017–2019

- **Network Anomaly & Fraud Detection:** unsupervised ML on streaming telemetry detecting anomalous traffic, DDoS signatures, and fraudulent behavior across high-volume distribution nodes.
- **Secure API & Data Architecture:** specs + access-control for the GRAPA API facade enabling secure authenticated retrieval from the enterprise data lake (HUB DATA France).
- **Threat Telemetry Pipelines:** Python ingestion parsing XML network logs into Elasticsearch / Apache Kudu as a tamper-proof repository for real-time security auditing.
- **Data Governance:** internal R package interfacing Impala / Hadoop, standardizing secure data-access protocols across the data-science org.

SELECTED PROJECTS

- **RepoSage — Agentic, execution-verified RAG** (LangGraph, vLLM, Qdrant, DPO): dual-source code-&-docs RAG that generates, executes in a sandbox, and self-corrects against a library's own tests; DPO alignment from execution-feedback preferences.
- **CipherGuard — reference-free backdoor detection** (GAT + AST, Isabelle/HOL-verified corpus): catches test-vector-compliant backdoors with zero false positives on 44 ciphers (NDSS 2027).
- **LLM Deception & Honesty Probing:** representation-engineering probes across Qwen3 (0.6B–32B); found the probe keyed on persona/chat-template tokens rather than content, and proposed incentive-based training.

EDUCATION

- **Ph.D., Computer Science** — Oakland University, Michigan (Sep 2020 – Apr 2026; GPA 3.96). Dissertation: *From Perceived Trust to Provable Security: Design, Optimization & Verification of Resource-Constrained IoT Ecosystems.*
- **M.Sc. Computer Engineering & B.Sc. Electronic & Computer Engineering** — ESIGELEC, France (2016–2019; GPA 3.87). *Big Data for Digital Transformation.*

SELECTED PUBLICATIONS

- V. Younang et al. "Security Risk Assessment with Bayesian Attack Graphs and Complex Probabilities." *IEEE TDSC*, 2025.
- V. Younang et al. "Extending IoT Devices' Network Lifetime." *IEEE TSC*, 2025.
- V. Younang et al. "Comparative Analysis of Users' Concerns and Government Policies on Autonomous Vehicles." *Springer ICIOT / LNCS*, 2022.
- V. Younang et al. "CipherGuard: Reference-Free Detection of Backdoors in Cipher Implementations." *Under review, NDSS 2027.*
- V. Younang et al. "Tiered Isomorphic Alignment: Curriculum Fine-Tuning across Six LLMs." *Under review, AAAI 2027.*
- V. Younang et al. "CryptoGraph: Learning Security Properties of Block Ciphers with Graph ML." *In prep (resubmission), IEEE TIFS.*

CERTIFICATIONS

- **AWS Certified AI Practitioner (AIF-C01)** — expected Nov 2026; AWS AI/ML stack incl. Amazon Bedrock and SageMaker.
- **NVIDIA-Certified Associate: Generative AI & LLMs (NCA-GENL)** — expected Dec 2026; transformers, prompt engineering, fine-tuning, NeMo / Triton / TensorRT.
- **OWASP Top 10 for LLM Applications** — de-facto LLM-security framework (prompt injection, insecure output handling, model DoS, supply-chain), demonstrated in RepoSage.